

P34 and PARML: Learning When Not to Trade

Reconstructing Market Signal from Biased Business Data

A generalized Profit-As-Regression decision model for biased, partially observed business markets

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Status and disclosure

This integrated paper consolidates the P34 industry release candidate and the P34 technical report into one publishable technical paper. It is not investment advice. It does not disclose deployment-sensitive architecture, training compute, private data, private connectors, trading endpoints, or exact live-deployment control logic. P34 should be evaluated through customer-specific backtests, synthetic stress tests, live shadow deployments, and staged risk controls before operational use.

Keywords: Profit-As-Regression Machine Learning; PARML; P34; business menus; selection bias; partial observability; false positives; portfolio construction; computable markets; latent-space recovery.

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Abstract

P34 is HyperC's implementation of Profit-As-Regression Machine Learning, or PARML: a model class that trains directly against final economic outcomes instead of relying on a chain of intermediate forecasts, dashboards, rules, and manual approvals. P34 is designed for computable markets in which the true market signal is not published, standardized, or cleanly observable, but where a business can observe structured opportunity telemetry, make repeated decisions, measure outcomes, and deploy a model into a live or semi-live decision loop.

The model consumes historical business menus and current inference-time menus. A business menu is a table representation of actions available to the business at a decision point: individual instruments, deal quantities, sub-portfolios, full portfolios, timing choices, derivative-like choices, or other market-compatible options. P34 learns from historical business activity, partially observed outcomes, safe sizing groundings, and the historical decisions made by a biased prior business policy. It is intended to decide not only what to trade, but also when the correct action is to refuse the trade.

P34 detects business and market regimes, builds nonparametric profit estimators, unrolls candidate portfolios, constructs multiple plausible interpretations of unobserved data, filters dominated interpretations, extrapolates them under possible drift, and selects a universe and portfolio according to a policy objective such as expected profit, downside protection, or false-positive control. Current nonparametric components use boosted models over materialized matrices in several blocks, but this is an implementation choice rather than the final intended limit of the architecture.

Synthetic benchmark evidence supports the intended behavior. In a declining-margin, periodic, hidden-bias benchmark, a tuned gradient-boosted regressor trained on fully grounded sizing options predicted +\$126.6k but realized approximately -\$219.8k. P34 predicted +\$20.7k and realized +\$14.8k, with 71.7% aggregate prediction calibration and 65% positive-portfolio success. In a stationary or growing-margin benchmark, P34 preserved performance comparable to the optimistic baseline, realizing +\$228.6k versus +\$227.1k for the baseline. These results do not guarantee live-market performance, but they demonstrate a controlled response to the failure mode P34 targets: optimistic false positives from biased business history under drift and partial observability.

Executive Summary

P34 is a decision model, not a price forecaster. Its output is an executable portfolio: selected trades and volumes. Its input is a business menu: a structured set of candidate actions that a business can actually take. The model class is PARML because the target is final economic outcome, normally portfolio-level profit, rather than a sequence of intermediate predictions.

- Core claim. P34 is designed to learn profitable portfolio selection from biased historical business data where only prior accepted trades have reliable outcomes.
- Operational goal. The system is optimized to avoid entering negative-expectation portfolios in difficult markets and to behave similarly to optimistic predictive systems when market conditions are stationary and predictable.
- Distinctive capability. P34 scans large menus of potential options and waits for favorable opportunities instead of focusing on one high-volume process or single liquid instrument.
- Business interface. The business provides data tables, action schemas, constraints, and outcomes; a business menu manager can maintain the menu without deep machine-learning specialization.
- Engineering philosophy. The architecture exposes clear layers: grounding, regime detection, base prediction, false-positive correction, universe construction, universe selection, and portfolio execution.
- Current implementation. Several nonparametric components use boosted models over materialized matrices; the intended architecture remains modular and implementation-agnostic.
- Evaluation result. In the principal difficult synthetic benchmark, P34 remained profitable while a tuned GBDT-R baseline lost approximately \$219.8k.

The strongest release claim is not that P34 forecasts every outcome better than existing models. The claim is that P34 is structured around the right economic problem: constructing portfolios from biased, incomplete business history while explicitly accounting for false positives, market regimes, and deployment-time uncertainty.

1. Introduction

Modern businesses increasingly operate in environments where the number of available decisions, data streams, constraints, counterparties, and market signals exceeds the capacity of traditional analytical systems. Classical business intelligence and forecasting pipelines separate the problem into several human-designed layers: data collection, dashboards, forecasts, rule construction, approval logic, and operational execution. Such systems are useful in stable and low-dimensional settings, but they become fragile when markets change, the feature space grows, or decisions must be made continuously at scale.

PARML reframes this problem. A PARML model is trained to estimate the economic result of a decision or portfolio directly. In the trading setting studied here, the target is portfolio-level profit. In adjacent domains, the same formulation can target recovery rate, contribution margin, liquidation value, risk-adjusted yield, utilization-adjusted revenue, or another final business objective expressed in economic units.

The central argument for PARML is that optimizing against final business outcomes connects telemetry to operations end-to-end. Instead of requiring a rigid stack of analytics, business rules, and manual decision-making, the model learns the relationship between high-dimensional business context and realized economic performance. New market signals, product attributes, counterparty features, operational constraints, and external data streams can be added without redesigning the decision system from first principles.

This shift has implications for enterprise value. A business whose decisions are encoded in spreadsheets, manual judgment, and brittle rules is difficult to scale, audit, and transfer. A business whose decision process is represented by a retrainable profit-directed model can become more stable, more observable, and more adaptive. In this framing, the model is not merely a predictive tool; it is an operational asset that converts business complexity into repeatable decision quality.

A second motivation is that learning final profit forces the model to internalize risk. Realized profit reflects price, timing, execution risk, capital constraints, inventory risk, liquidation risk, market competition, hidden costs, adverse selection, and operational friction. A sufficiently accurate PARML model must therefore learn practical regularities of the business process. Stable learned relationships between telemetry and economic outcomes are referred to here as emergent business physics. These relationships are empirical and must be validated, but they can be probed by controlled perturbations, holdouts, forward tests, and live feedback.

P34 differs from financial time-series foundation models such as Kronos. Kronos pretrains on financial K-line records to support forecasting, volatility prediction, and synthetic K-line generation. P34 is not primarily a K-line sequence model. It is a business decision layer that consumes a configurable action menu and optimizes final portfolio economics under partial observability.

2. Scope, Positioning, and Report Boundaries

This paper focuses on the capabilities, limitations, testing methodology, and safety properties of P34. P34 treats business decision-making as a direct optimization problem over future economic outcomes rather than a sequence of intermediate prediction tasks. It learns from historical business telemetry to estimate the profitability of alternative portfolios across multiple plausible interpretations of partially observed market behavior.

For competitive and safety reasons, this paper does not disclose deployment-sensitive implementation details such as exact model sizes, private feature inventories, production hardware, training compute, serialized selector models, proprietary data, private connectors, trading endpoints, or exact live-deployment control logic. It describes the model class, interface, data assumptions, evaluation design, benchmark results, limitations, governance requirements, and market implications at a level appropriate for industry and technical review.

The current implementation uses boosted and other nonparametric components over heavily materialized matrices in several blocks. These components are a practical debugging and iteration choice. The architectural intent is a lifted, modular decision system that can eventually be implemented as a more differentiable neural architecture with full backpropagation across a broader signal flow.

The evidence base consists of internal model design, inspection of supplied synthetic-market notebooks, and inspection of supplied backtest notebooks. The benchmark results should be read as technical-report evidence rather than a claim of

broad market dominance. They demonstrate behavior in controlled synthetic regimes deliberately built to expose real-world partial-observability and selection-bias challenges.

P34 is currently presented as an early technical release. Operational use should proceed through customer-specific backtests, live shadow deployments, staged capital exposure, monitored pilots, hard risk limits, and rollback procedures.

3. The Core Problem: Biased Business Data Creates Optimistic Models

Most businesses do not observe the full market. They observe the deals that appeared in their systems, and they know reliable outcomes mainly for the deals they accepted. Those accepted deals were not random: they were selected by human operators, rules, or prior models trying to avoid unprofitable opportunities. The resulting historical dataset is therefore selection-biased toward best-effort profitable trades.

A conventional supervised model can look strong on holdout tests drawn from the same biased history and still fail at deployment time. The backtest may appear accurate because it tests on optimistic historical data. At production time, the model faces a wider opportunity set and many more false positives: cases that look profitable under the historical pattern but realize negative profit. In the extreme, this can produce negative ROI even when the historical backtest looked good.

This is the failure mode P34 is designed to address. Its objective is not only to predict profits, but to construct portfolios that remain economically useful when the market is noisy, partially observed, and drifting. In low-margin businesses, avoiding false-positive trades can be as important as finding true positives.

The no-trade decision is therefore a core output, not an afterthought. P34 should be rewarded for waiting when the expected portfolio outcome is negative or uncertain beyond policy limits. In low-margin markets, the ability to refuse a trade can be more valuable than the ability to rank every available opportunity.

4. Computable Markets, Latent-Space Speculation, and Applicability Scope

P34 is intended for computable markets. A computable market is a market where a model can observe sufficient structured telemetry, generate a business-compatible action menu, make repeated decisions, receive measurable economic feedback, and be deployed into a live or semi-live decision loop. The market does not need to be perfectly efficient, fully digitized, or high-frequency, but it must be measurable and repeatable enough for model improvement to be evaluated.

From a classical finance point of view, P34 targets market inefficiencies created by incomplete or unavailable market data. Many markets outside public securities do not have a Bloomberg Terminal-like view of order books, market depth, transaction volume, counterparty behavior, or true inventory availability. Retail is a clear example: precise information about where inventory sits is often hard or impossible to observe because inventory may be in transit, in processing, in micro-batch production, or locked behind operational delays.

Additional inefficiency comes from weaker standardization, less formal reporting, non-uniform participant behavior, operational instability, financial-control differences, and scaling limits inside the enterprises executing trades. These factors make the explicit market signal unavailable, but they do not make the underlying economics disappear. The same mechanics often exist in latent form across operational data, business processes, payment behavior, lead quality, fulfillment timing, inventory movements, and other modalities.

P34 is designed to exploit this latent signal. The industry article calls this frontier latent-space speculation: estimating economic value in markets where the true market state exists but is not explicitly materialized as a clean market data feed. The broader recovery process is called latent-space recovery: creating a rigorous financial dashboard and decision infrastructure for businesses that previously struggled to reconcile operational performance, estimate asset value, and access financial vehicles that require predictable cash-flow evidence.

A crypto example makes the point concrete. Suppose an operator wants to trade \$100,000 of a newly listed token, LOLQD, where the largest prior trade in a bullish market was only \$10,000. Historical data at the smaller scale does not describe the true market impact, liquidity, or execution risk of the intended trade. The same optimistic curse appears in Amazon wholesale when a business has only traded small quantities and only records the deals it previously believed were good. P34 is designed for these cases: estimate portfolio value where the explicit historical sample is incomplete and biased, but the latent operational signal may still be learnable.

The opportunity is large but bounded. Public retail estimates frequently place worldwide retail sales in the low tens of trillions of dollars annually, and the source article cites a National Retail Federation forecast reported by AP News that U.S. retail sales would reach approximately \$5.6 trillion in 2026. P34 does not claim to address all retail or all commerce at once; it targets the subset of markets where menus, repeated decisions, measurable outcomes, and sufficient telemetry are present. Exact market-size estimates should be source-checked before use in external fundraising material.

Table 1. Candidate conditions for a computable PARML market.

Condition	Interpretation
Representable actions	Available actions can be represented as rows in a table, including candidate instruments, deal quantities, prices, timing, counterparties, or portfolio choices.
Measurable outcomes	Final economic outcomes can be measured after an acceptable delay, including revenues, costs, defaults, returns, liquidations, writeoffs, and operating expenses.
Historical policy signal	The business can provide enough historical decisions to estimate how prior policy bias shaped the observed data.
Executable constraints	The model can be constrained by business rules such as budget, risk, inventory, compliance, capital, and operational capacity.
Limited or modeled impact	The action has limited or measurable market impact; if taking one action destroys the remaining menu in an unobservable way, additional modeling is required.
Testability	Holdout, forward, shadow, or live evaluation is possible without exposing the business to uncontrolled downside.

Examples of potentially computable markets include retail and wholesale trading, GPU capacity allocation, apparel resale, liquidation markets, electricity markets, moving leads, trucking load boards, freight consolidation, collectible goods, call markets, payment transaction routing, used vehicles, specialty chemicals, short-duration fixed income, tax and environmental credits, prediction markets, digital game assets, construction materials, and newly created options-like markets around currently illiquid goods or services. This list is illustrative rather than exhaustive.

A market may not be computable if actions change future availability in ways the model cannot observe, if the feedback delay is too long or too noisy, if outcomes cannot be attributed to decisions, if legal or compliance constraints prevent automation, or if market impact dominates expected profit. In such environments, the appropriate first step is instrumentation: telemetry capture, menu design, outcome measurement, and shadow testing.

5. Business Menus, Context, and the Input/Output Contract

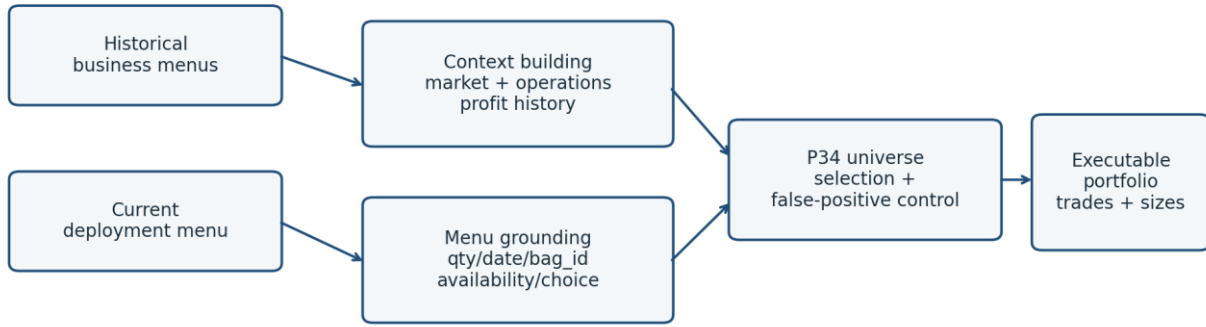
The main operational concept in P34 is the business menu. A business menu is the set of options the business can choose from at a decision time. Historically, the menu contains the full set of options the business could have chosen from, but outcomes are only partially known because the business chose only some options. At deployment time, the menu is the current opportunity set from which P34 constructs an executable portfolio.

Menus can be daily menus, vendor-level menus, lead-channel menus, load-board menus, instrument menus, or portfolio-level menus. The menu can contain direct instruments, derived instruments, sub-portfolios, complete portfolios, timing choices, and sizing choices. The model constructs a portfolio from selected menu options and predicts portfolio-wise profit while accounting for risk. The portfolio may include direct instruments, sub-portfolios, whole portfolios, timing choices, derivative-like instruments, or market-specific options.

Table 2. Business-menu special columns.

Special column	Meaning
qty	Sizing of the option: units to order, loan-size step, contract size, number of coins, or another discrete/continuous quantity.
date	Historical order of menus or deployment-time ordering context.
bag_id	Logical separation of menus, for example multiple menus traded on the same day, by vendor, by channel, or by execution event.
menu_available	Binary indicator showing whether the option quantity was actually available for this business to choose, or only observed as historically present elsewhere in the market.
historically_chosen	Binary indicator showing which option the business actually took historically.

P34 menu-to-portfolio operating concept



Business menu special columns: qty, date, bag_id, menu_available, historically_chosen

Goal: estimate portfolio-wise profit under partial observability and choose the actions worth executing.

Figure 1. P34 menu-to-portfolio operating concept. Historical business menus and current deployment menus are grounded into a context and menu representation; P34 then performs universe selection and false-positive control to output executable portfolio trades and sizes.

The menu abstraction is designed to make P34 attainable for normal business IT teams. A business does not need to specify neural-network internals. It needs to provide well-formed tables that describe actions, keys, dates, menu identifiers, candidate sizes, available telemetry, historical choices, and outcomes. HyperC can supply rule-based market connectors that translate common operational telemetry into menus for specific industries, such as wholesale trade or credit-like workflows.

A useful organizational role is the business menu manager. This role can be paired with database administration, business architecture, or data modeling. Its responsibility is to understand operational rules, maintain menu schemas, validate data integrity, and ensure that the menu remains business-compatible as markets, vendors, products, and constraints change.

The word context maps to an intuition people already have from large language models: context is the task definition plus necessary priors. In P34, there are two distinct contexts.

1. Market-and-business context. Historical menus and realized profits define the market access, business operating capability, financial controls, scaling limits, and typical market conditions. This context requires representation learning through model fitting.
2. Deployment-time menu context. The current opportunity set represents the business options available now, together with the current market state. This context is evaluated to produce the next portfolio decision.

For more standardized markets, the context update interval can be longer. In an Amazon wholesale-like setting, a market context may be provided once and reused across many businesses if their operations are sufficiently standardized and differ mainly by the inventory they can access.

P34 assumes that real business data is biased and partially observed. A business rarely sees the outcome of every trade it could have entered, because it usually takes only the trades it believes are profitable. This creates a logged-data problem: positive examples are overrepresented, unobserved actions have missing outcomes, and the prior decision policy is unknown or only partially known.

The business menu approach allows additional safe grounding. In some markets, if the business executed a transaction of size N , it may be safe to infer outcomes for smaller quantities $m < N$ under the same batch or deal context. For example, in a wholesale resale setting, if N items were sold, the outcome of selling the first m items can often be grounded more safely than the outcome of selling more than N items. P34 benefits from these safe label expansions, but the architecture does not require full market labels.

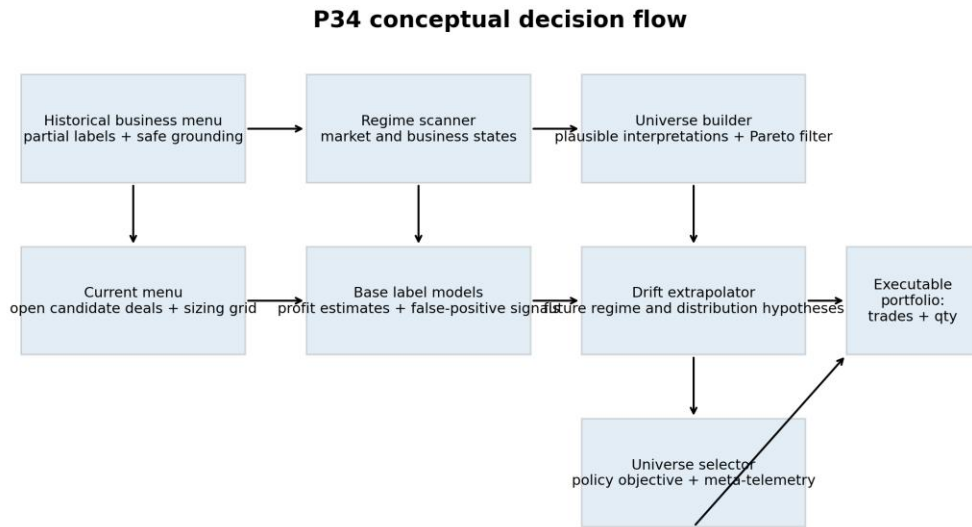
Table 3. Simplified P34 input contract.

Input family	Meaning
X_grounded_biased	Historical business telemetry for labeled or safely grounded actions.
y_grounded_biased	Observed economic outcome, normally profit, for grounded historical rows.
x_grounded_biased_keys	Deal, instrument, ASIN, lead, or other mutual-exclusion key.
x_grounded_biased_dates	Decision or market time. Used for regimes, holdouts, and drift tests.
x_grounded_biased_bag_ids	Business menu, portfolio, vendor, or market-access grouping.
x_historical_biased_choices	Actions selected by the historical business policy, used to estimate selection bias.
X_grounded_deployment_unlabeled	Current or historical unlabeled candidate actions.
x_grounded_deployment_unlabeled_*	Keys, dates, and bag identifiers for current or candidate menus.

At inference time P34 receives a current menu with less history than the training context. The menu can be a sliding window, a current selection of possible trades, or a fixed portfolio candidate set. Volumes do not need to be supplied as final decisions; candidate volumes can be unrolled as options, and P34 outputs selected trades and volumes as a portfolio.

6. P34 Model Architecture

P34 is a generalized decision model that learns from partially observed business histories to construct latent representations of market regimes, business dynamics, and decision outcomes. These representations are used to evaluate and optimize portfolios directly against economic objectives such as profit, yield, recovery rate, or contribution margin.



The current implementation uses nonparametric boosted-model layers in several blocks; the architectural intent is implementation-agnostic and compatible with a future differentiable stack.

Figure 2. Conceptual P34 decision flow. The diagram intentionally omits deployment-sensitive implementation details. The current implementation uses nonparametric boosted-model layers in several blocks; the architectural intent is implementation-agnostic and compatible with a future differentiable stack.

6.1 Regimes, Labels, and Universes

P34 scans historical business activity for market and business regime changes. A regime may represent a change in market difficulty, margin compression, acceptance policy, liquidity, inventory behavior, demand quality, or external signal reliability. The model names and represents these regimes so later components can decide which interpretation of the business history is currently applicable.

The model then builds nonparametric label predictors for profit and related false-positive telemetry. In the current implementation, boosting-based models are used for several of these predictors. This is a practical implementation status, not a claim that the architecture must remain boosting-based.

Because the training data is partially observed, P34 does not assume there is one complete ground-truth interpretation of the historical market. It constructs multiple plausible universes. A universe is a candidate interpretation of the historical data, including assumptions about selection bias, unobserved opportunities, false-positive risk, market regime, and future drift. A Pareto filter removes interpretations that are dominated under the configured policy objective.

6.2 False-Positive Correction and Controlled Speculation

A major failure mode in low-margin markets is the false positive: a trade appears profitable but carries hidden downside such as dead stock, liquidation risk, execution failure, default, or margin compression. P34 therefore treats false-positive correction as a first-class modeling target. It unrolls candidate portfolios and predictions into matrices that expose where a selected deal would produce apparent profit but unacceptable realized downside.

The model also injects controlled speculation into the meta-level. This is not arbitrary optimism. It is a structured way to account for the business desire to take more volume, the presence of unseen profitable cases, and the fact that the prior business policy may have been conservative. Speculative interpretations compete with conservative interpretations and are selected only if the universe selector judges them consistent with generalized telemetry and the configured policy.

6.3 Universe Selector

The universe selector chooses among plausible universes using generalized telemetry. Its inputs are intentionally meta-level signals: regime features, market telemetry, portfolio prediction quality, false-positive correction telemetry, and signals that summarize the behavior of candidate portfolios. Because the selector does not rely only on specific market identities, it is more reusable than a narrow market-specific regressor.

The selector can optimize different definitions of correctness. In some deployments, correctness means minimizing downside risk or spread. In others, it means selecting the universe with the most accurate portfolio-profit prediction. In more aggressive settings, it may optimize a profit-weighted false-positive metric. Human-supplied or LLM-supplied market laws can be injected as structured signal flows, while nonparametric layers fine-tune the latent variable space.

6.4 Lifted Labels

P34 uses lifted-label signals to reduce dependence on fully synthetic market labels. Instead of training only against labels from a fabricated full market, the model can use holdouts, forward windows, out-of-fold predictions, portfolio-level calibration, and false-positive telemetry to create realistic performance signals. In the synthetic benchmarks, full future outcomes are available for evaluation, but the training path is designed to resemble the partial feedback available in real business operations.

7. Engineering Design Principles

The model was designed around a practical engineering constraint: business-finance decision systems are harder for engineers to reason about than text, images, or conventional prediction tasks. P34 therefore exposes concrete engineering surfaces. Each surface has a simple improvement question: is the menu grounded correctly, are regimes detected, are false positives corrected, is the right universe selected, is sizing stable, and are external signals improving decisions?

7.1 Multi-Lifted Modular Architecture

The architecture is multi-lifted in the sense that signals can be passed upward from raw telemetry to regimes, from regimes to label predictors, from label predictors to portfolio universes, and from universes to executable policies. Engineers can add compute and signal flow at well-defined points without redesigning the entire model. More compute can expand candidate universes, improve feature selection, increase portfolio unrolling depth, or refine selector telemetry.

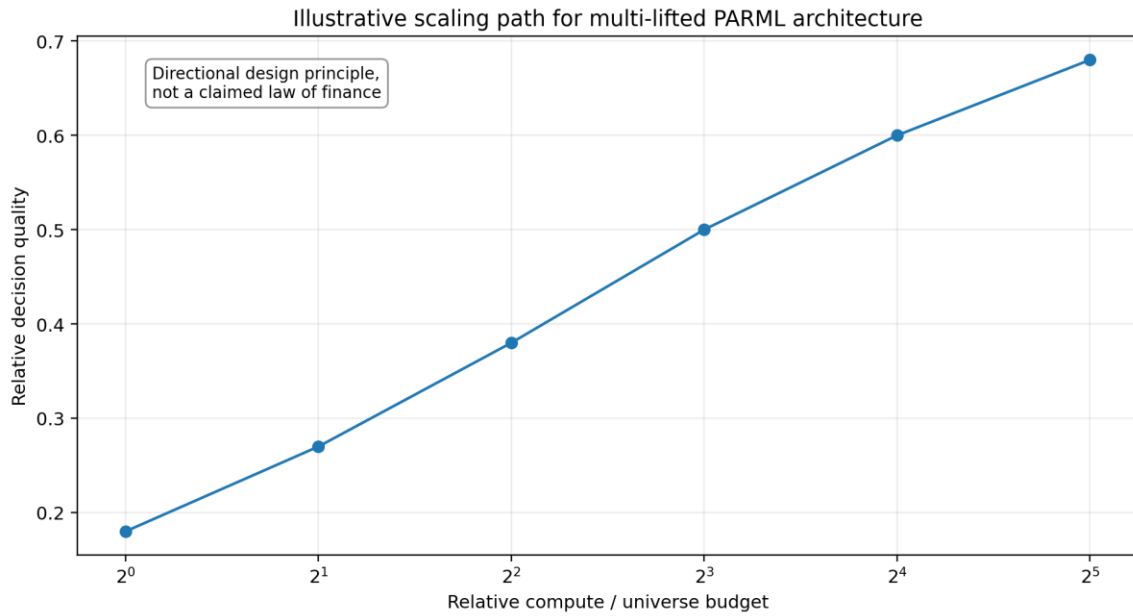


Figure 3. Conceptual scaling path for a multi-lifted PARML architecture. The figure illustrates a design principle; it is not a universal law or a financial guarantee.

7.2 Broad Feature Ingestion with Auto-Pruning

P34 uses a principle internally called careless feature engineering: the system can ingest broad feature sets while relying on stage-level pruning, feature selection, and validation to remove harmful or non-generalizing signals. This is designed to make the model resilient to new data streams and to reduce the risk that a business must manually curate every feature before experimentation.

Careless feature engineering does not mean careless data management. Data provenance, leakage checks, date-aware splits, feature stability, missingness, and operational meaning remain important. The principle is that the model architecture should absorb candidate signals quickly and then reject or downweight them when they do not improve portfolio-level objectives.

7.3 Boundary-Aware Sizing

Deal sizing is a first-class output. P34 must handle zero quantity, one unit, small quantities, and large discrete volumes. Boundary effects are important because profit curves are often non-monotonic: buying more units may increase gross margin up to a point and then create inventory or liquidation loss. The model therefore treats size as part of the action space rather than as a continuous afterthought.

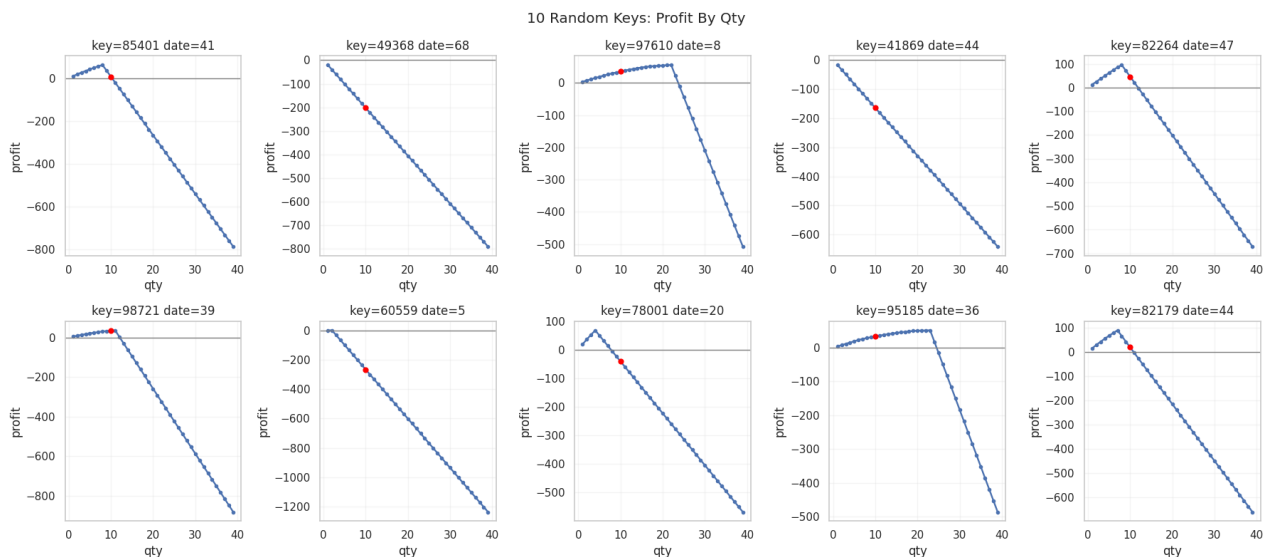


Figure 4. Example synthetic profit curves by quantity. The curves illustrate why deal sizing cannot be treated as a simple monotonic scalar.

7.4 External Signal Reuse

P34 is designed to reuse signals from other markets. A fine-tuned market indicator, a demand forecast, a counterparty quality score, or a futures-derived signal can be added to the business menu and evaluated in the same profit-directed framework. This enables P34 to compete with systems that use only market-specific internal data by learning which external signals matter for a specific menu request.

7.5 Human-Understandable Debugging

A central goal is to make engineering work straightforward. When a P34 portfolio fails, the question should not be an opaque "why did the model choose that?" Instead, engineers should be able to ask:

- Was the action menu incomplete or improperly grounded?
- Did the system detect the wrong market or business regime?
- Did the base profit model overestimate the label?
- Did false-positive correction fail?
- Did the universe selector choose a risky interpretation?
- Was the sizing boundary wrong?
- Did an external signal drift or become adversarial?
- Was the portfolio constrained incorrectly by business rules?

8. Synthetic Market and Testing Methodology

The supplied market-synthesis notebook builds synthetic markets that are intentionally simple yet expose the difficulties of real observational business data. The goal is not to reproduce every detail of a real market. The goal is to build a controlled environment where selection bias, partial observability, hidden business policy, regime variation, downside shocks, and distribution drift can be isolated and tested.

8.1 Synthetic Market Construction

The inspection configuration generated 3,931,200 telemetry points. The split used 80 historical dates and 20 future evaluation dates, with the future split beginning at date 80 and the historical split ending at date 79. The main inspection bundle contained 3,024,450 historical rows and 906,750 future-evaluation rows. Each candidate deal was unrolled across 39 possible quantities.

The generator uses latent demand archetypes, cost draws, weekly demand paths, market difficulty cycles, market trend, hidden latent market drivers, downside regimes, and a biased business acceptance policy. The historical business policy accepts a non-random subset of possible offers, with a small amount of exploration. In the inspection run, policy exploration accepted 166 of 16,138 accepted-by-policy deals, or approximately 1.03%, matching the configured 1% exploration target.

Table 4. Key metrics from the supplied market-synthesis notebook inspection run.

Synthetic-market metric	Inspection value
Telemetry points generated	3,931,200
Historical rows	3,024,450
Future evaluation rows	906,750
Historical dates	80
Future evaluation dates	20
Labeled rows after safe grounding	528,060
Unlabeled historical rows	2,496,390
Observed-label share after grounding	17.46%
Quantity values per key	39
Labeled positive-profit rate	19.23%
Unlabeled positive-profit rate	2.58%

8.2 Bias and Partial Observability

The benchmark deliberately makes labeled data different from unlabeled data. In the inspection run, the labeled set had higher average margin and higher positive-profit rates than the unlabeled set. This reproduces the central real-world problem: businesses do not randomly label all possible actions; they label the actions selected by their prior decision policy and sometimes smaller safe groundings.

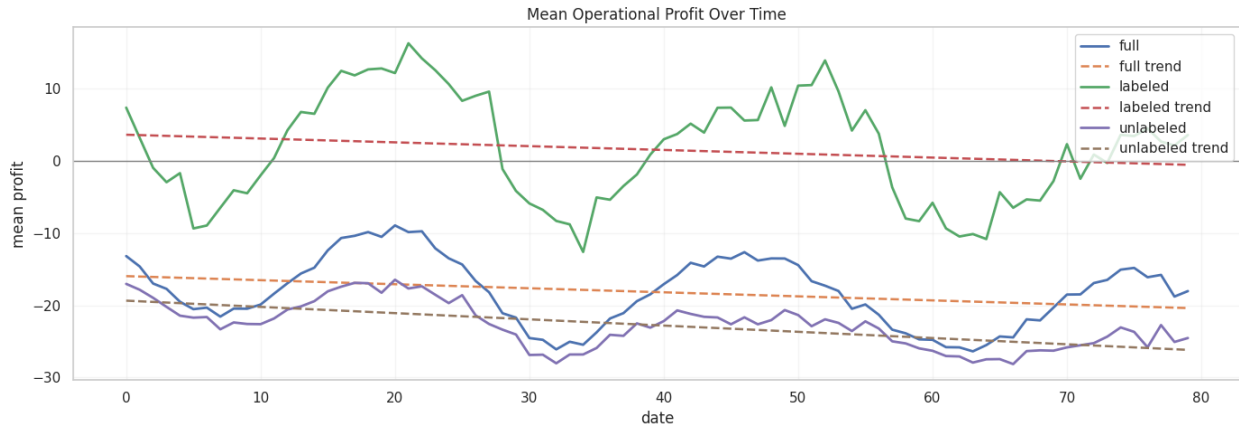


Figure 5. Mean operational profit over time in the synthetic market. Labeled and unlabeled segments have different distributions and trends, illustrating selection bias.

The synthetic benchmark also validates that acceptance-rate variation is driven by both market difficulty and hidden business policy. A diagnostic in the notebook found that market difficulty alone explained $R^2 = 0.835$ of observed acceptance-rate variation, hidden policy threshold alone explained $R^2 = 0.122$, and both together explained $R^2 = 0.951$. The approximate explained-variance decomposition assigned 87.5% to market difficulty and 12.5% to latent business rule changes.

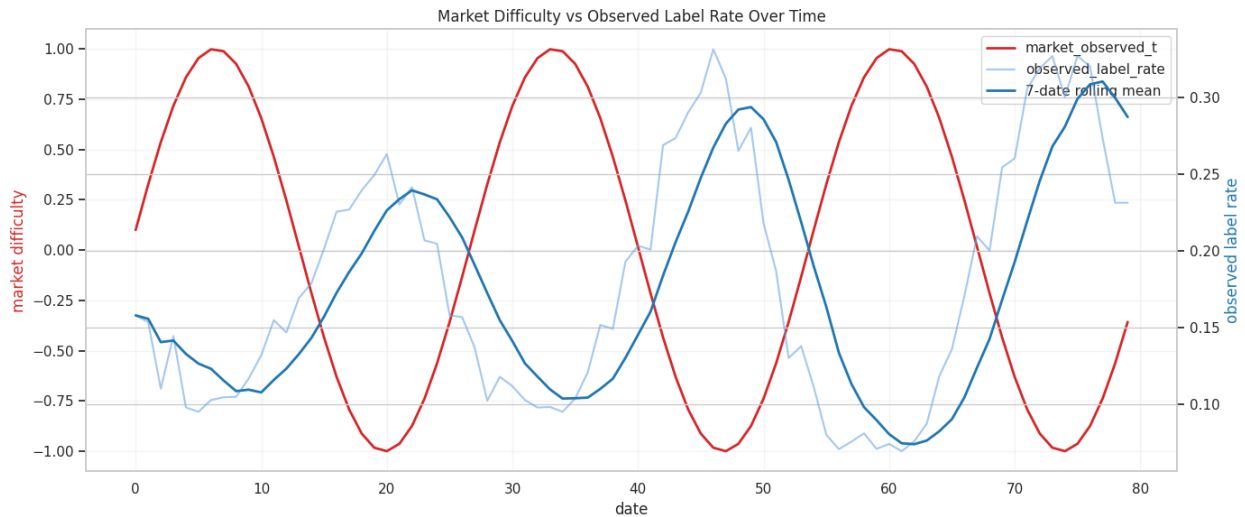


Figure 6. Market difficulty and observed label rate over time. The observed label process is not stationary and not random.

8.3 Baseline System

The principal baseline is a state-of-art business-grounded gradient-boosted decision-tree regressor, abbreviated GBDT-R in this paper. It is trained on fully grounded sizing options with tuned parameters and Huber-style robust regression behavior. At evaluation time, it predicts profit for candidate rows and selects the best positive predicted quantity for each key. This approximates what a capable business data-science team might build with standard tabular ML tools and a hand-written portfolio-selection wrapper.

The baseline is intentionally strong in the stationary setting because it has access to well-grounded row-level features and quantity options. Its failure in the difficult setting is therefore informative: the issue is not lack of a powerful

regressor, but a mismatch between row-level prediction, biased observation, false-positive risk, and portfolio-level profitability under drift.

8.4 Backtest Mechanics

The backtest treats each date or bag as an inference-time business menu. For each bag, both the baseline and P34 evaluate candidate quantity rows. P34 returns selected key-quantity rows as a portfolio. The notebook then joins selected rows back to the fully known synthetic evaluation target to compute realized profit. It also computes an oracle positive upper bound: for each key, choose the true best quantity if its profit is positive, and otherwise choose nothing.

The backtest notebook stores per-bag multiverse telemetry for post-hoc inspection. This telemetry includes predicted profit, actual profit, multiverse index, false-positive correction, and Pareto-universe features. This design supports the engineering goal of diagnosing where a decision failed rather than simply reporting aggregate profit.

8.5 Evaluation Metrics

Table 5. Evaluation metrics used in the report.

Metric	Definition
Realized portfolio profit	Sum of actual profit for selected key-quantity rows across evaluation bags.
Predicted portfolio profit	Sum of predicted profit for selected rows.
Prediction calibration	Actual selected profit divided by predicted selected profit, reported as a percentage when predicted profit is positive.
Positive-portfolio success rate	Share of evaluation bags where the selected portfolio realized positive profit.
Oracle positive upper bound	Sum over keys of $\max(0, \text{best actual quantity profit})$, not available in deployment.
Market efficiency	P34 realized profit divided by oracle positive upper bound.

9. Performance Benchmarking

No directly comparable publicly available model design was available for a like-for-like benchmark. The report therefore compares P34 against a strong standard tabular ML approach operating on the same synthetic business menus. The evaluation focuses on two contrasting conditions: a difficult declining-margin environment with hidden bias and a more favorable stationary/growing environment.

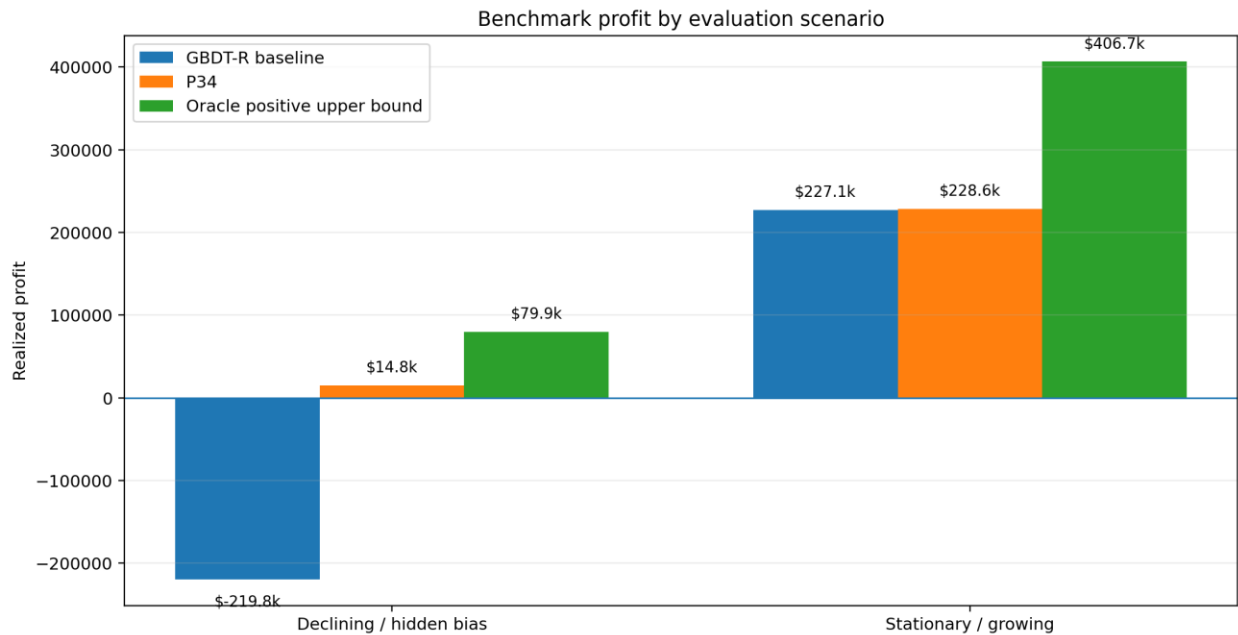


Figure 7. Benchmark profit comparison. The oracle positive upper bound is not deployable; it is included to contextualize efficiency.

Table 6. Principal benchmark results: actual and predicted portfolio profit.

Scenario	GBDT-R actual	GBDT-R predicted	P34 actual	P34 predicted
Declining margin, periodic hidden bias	-\$219,805.62	+\$126,597.70	+\$14,814.50	+\$20,662.15

Scenario	GBDT-R actual	GBDT-R predicted	P34 actual	P34 predicted
Stationary / growing market	+\$227,051.73	+\$287,794.31	+\$228,605.88	+\$377,959.86

Table 7. Principal benchmark results: calibration, success, oracle upper bound, and efficiency.

Scenario	P34 calibration	P34 positive portfolios	Oracle positive	P34 efficiency
Declining margin, periodic hidden bias	71.70%	65.00%	\$79,893.67	18.54%
Stationary / growing market	60.48%	90.00%	\$406,656.24	56.22%

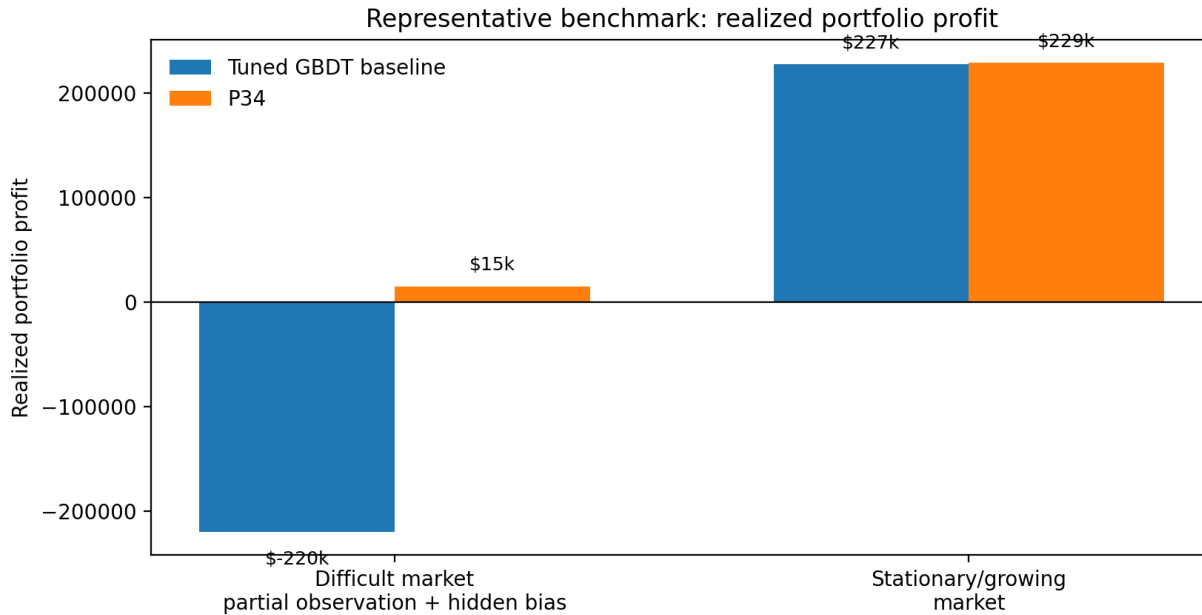


Figure 8. Industry-facing representative benchmark: P34 remains positive in a difficult market where a tuned baseline realizes a large loss, while remaining competitive in a stationary/growing market.

Table 8. Industry-facing benchmark interpretation.

Scenario	Conventional baseline	P34	Interpretation
Difficult market	Tuned GBDT predicts +\$126.6k but realizes -\$219.8k.	P34 predicts +\$20.7k and realizes +\$14.8k.	Tests whether P34 can avoid the optimistic false-positive collapse mode.
Stationary/growing market	Tuned GBDT realizes +\$227.1k.	P34 realizes +\$228.6k.	Tests whether P34 preserves upside when ordinary forecasting works.

9.1 Difficult Market: Declining Margin and Hidden Bias

In the difficult benchmark, the synthetic market contains periodic market difficulty, partial hidden business bias, and increasing market efficiency that compresses margin over time. This is the setting where naive profit regressors are most dangerous: they can extrapolate positive outcomes from historical accepted trades into a future market where the same pattern is no longer profitable.

The GBDT-R baseline predicted +\$126.6k of profit but realized -\$219.8k. This is a classic false-positive failure. The model entered trades that looked profitable under its learned row-level predictions but produced negative realized portfolio profit. P34 predicted +\$20.7k and realized +\$14.8k. Although its efficiency versus the oracle positive upper bound was only 18.54%, its primary success was avoiding the catastrophic negative portfolio chosen by the baseline.

9.2 Stationary/Growing Market

In the stationary/growing benchmark, market margin improves rather than compresses. This setting rewards optimistic row-level prediction because historical relationships remain more stable. The GBDT-R baseline realized +\$227.1k. P34 realized +\$228.6k, preserving comparable performance while selecting portfolios that were positive in 90% of evaluation bags. This result is important because a robust PARML architecture should not avoid risk so aggressively that it sacrifices performance in easy markets.

9.3 Interpretation

The combined result supports the intended model behavior. P34 is not simply a conservative filter. It behaves conservatively enough to avoid severe loss under drift and selection bias, while preserving upside when market conditions are stationary and predictable. The benchmark also shows why row-level predictive accuracy is insufficient: a model can predict positive aggregate profit and still select a loss-making portfolio if it fails to account for false positives, hidden prior-policy bias, and regime change.

The results do not prove universal profitability. They show that the architecture addresses a specific and common failure mode in real operational data: the business only observes actions its prior policy selected, and a standard regressor trained on these labels can over-enter trades when the market moves. P34 reduces this failure through universe construction, false-positive correction, and meta-level universe selection.

10. Competitive Landscape and Related Systems

P34 does not map cleanly to one existing software category. The closest substitutes are assembled stacks that combine portfolio analytics, quant research infrastructure, AutoML, decisioning platforms, optimization solvers, and custom engineering. P34 is best positioned as the missing decision layer between enterprise telemetry and executable economic action.

Table 9. Adjacent public categories and differences relative to P34.

Category	Representative systems	Difference relative to P34
Portfolio and risk platforms	BlackRock Aladdin, Bloomberg PORT, MSCI/Barra, FactSet	Strong in whole-portfolio analytics, risk, accounting, compliance, and operating workflows. They are not publicly positioned as profit-regression engines trained from biased business histories.
Quant research and trading infrastructure	QuantConnect, Alpaca, broker APIs, backtesting engines	Strong for research, backtesting, optimization, and live execution. They usually expect the user to design the strategy.
Decision intelligence and rules	SAS Intelligent Decisioning, Pega Customer Decision Hub, FICO, Palantir-style operational platforms	Strong for operational decision orchestration. Typically rule/model workflow systems, not arbitrary business-menu portfolio-profit learners.
AutoML and enterprise AI	DataRobot, H2O.ai, Dataiku, custom XGBoost/LightGBM stacks	Strong for predictive model development. They do not natively define the PARML target, handle partial observation, construct universes, size deals, and output executable portfolios.
Financial foundation models	Kronos, BloombergGPT, FinGPT, financial LLM copilots	Strong for financial time-series or language tasks. They can supply signals to P34 but do not themselves represent menu-based autonomous portfolio construction.
Academic policy learning	Contextual bandits, counterfactual risk minimization, offline RL, FinRL	Addresses logged feedback and policy learning, but practical business deployment still requires menu design, safety limits, partial-observation treatment, and operational integration.

10.1 Why P34 Is Not Just AutoML

AutoML platforms can train high-quality tabular models once the supervised-learning problem is defined. The harder problem in PARML is defining the economic learning problem: how to turn biased historical decisions into a portfolio-profit objective, how to represent unobserved alternatives, how to avoid false positives, how to size actions, and how to select robust portfolios. AutoML can be a component inside P34, but it is not the complete architecture.

10.2 Why P34 Is Not Just Offline Reinforcement Learning

Offline RL and contextual bandit research are conceptually close because they study policy learning from logged data. However, many algorithms require assumptions about exploration, propensities, or simulator coverage that real businesses often cannot provide. Offline RL is also vulnerable to extrapolation error when the learned policy chooses actions outside the distribution of the logged policy. P34 addresses this practical gap by explicitly constructing multiple plausible universes, filtering them, and selecting portfolios with false-positive correction and business constraints.

10.3 Why P34 Is Not Just a Trading Platform

Trading and backtesting platforms help teams implement strategies, test them, and execute them. P34 attempts to learn the decision and sizing policy itself from historical business menus. This is why the scope extends beyond listed securities. The same abstraction can apply to wholesale orders, freight bids, moving leads, GPU capacity, collectible goods, electricity, liquidation lots, credits, or other computable markets.

11. Safety, Risk, and Governance

A profit-directed autonomous decision model can create material economic risk if deployed without controls. P34 should be treated as a decision system with financial, operational, and compliance consequences. The appropriate deployment pattern is staged: offline backtest, synthetic stress test, shadow mode, limited capital exposure, monitored live pilots, and only then scaled deployment.

Table 10. Primary P34 deployment risks and mitigations.

Risk	Failure mode	Mitigation
False-positive over-entry	Model selects trades that appear profitable but carry hidden downside.	False-positive telemetry, downside policies, hard loss limits, universe selection, and shadow validation.
Distribution shift	Market regime changes after training.	Regime detection, drift extrapolation, live calibration, rollback rules, and no-trade defaults.
Market impact and liquidity	Taking an action changes the economics of other actions.	Liquidity constraints, menu impact modeling, concentration limits, and market-specific guardrails.
Data leakage	Future information or post-decision outcomes leak into training features.	Date-aware splits, feature provenance, leakage audits, and replayable pipelines.
Bad business menu	Menu misrepresents actual executable options or constraints.	Business menu manager, schema tests, rule validation, and reconciliation to execution systems.
Automation failure	API, execution, inventory, capital, or operational systems fail.	Kill switches, order throttles, reconciliation, manual override, and audit logs.
Regulatory or contractual risk	Automated decisions violate legal, platform, or counterparty constraints.	Policy constraints embedded in menus, compliance review, allowed-market list, and deployment approvals.
Adversarial behavior	Simpler market participants or counterparties react to the model.	Adversary tests, exposure limits, changing execution patterns, and live anomaly detection.

11.1 No-Trade as a Safety Action

A core safety property of P34 is that no-trade is an available action. The system should be rewarded for not entering a trade when the expected portfolio outcome is negative or uncertain beyond policy limits. This distinguishes P34 from systems that must always output a forecast or ranking. In low-margin markets, the ability to wait is often more valuable than the ability to predict every opportunity.

11.2 Independent Auditing

Independent auditing is important because backtests can be misleading, especially when the model, data, and evaluation are all developed by the same team. Appropriate audits include leakage review, split review, synthetic generator review, portfolio-reconstruction review, false-positive stress tests, execution-log reconciliation, and live shadow-mode validation. HyperC should provide controlled access to testing toolkits and APIs to qualified auditors while protecting deployment-sensitive details.

12. Market Implications: Hidden Telemetry to Financial Assets

The broader implication of P34 is not only better operational decisions. If a business can reconstruct enough latent market telemetry to make reliable predictions about non-security assets, then those predictions may become a basis for financial rigor and new liquidity. For example, a business with millions of menu options could estimate the expected trade-result values of issued purchase or manufacturing orders. If those predictions repeatedly prove reliable, the business may be able to resell or finance such orders in markets that require predictable returns.

This is analogous in spirit, not in mechanism, to the way standardized option-pricing frameworks helped organize options markets. A widely accepted benchmark for non-standard assets can increase confidence, standardize expectations, and unlock liquidity. P34 aims to create such benchmarks for computable operational markets where the asset is not traditionally considered a security but still has measurable expected economic value.

This is the latent-space recovery thesis: businesses can convert hidden telemetry into structured market signal, financial dashboards, auditability, and decision infrastructure. In markets that were previously too operationally messy for conventional analytics, this may allow asset-like valuation of purchase orders, manufacturing orders, lead streams, capacity blocks, or other measurable rights to future economic outcomes.

13. Example Application: Venture Investing as a Biased-Data Market

A venture investor provides a useful analogy for readers in the capital markets and technology ecosystem. The market may contain information about many investors funding many startups, but the dataset visible to a single Silicon Valley VC is biased. The investor sees a subset of deals filtered by network, geography, reputation, introductions, partner preferences, and prior selection behavior. The industry observes its own performance only after trying to enter selectively good startup deals. A model trained on such data may become optimistic for the same reason a retail-trading model becomes optimistic: the data reflects best-effort selection, not the full opportunity distribution.

P34 is not presented here as a finished VC underwriting product. The point is that venture investing illustrates the same mathematical problem: decisions under partial observability, biased historical labels, noisy outcomes, and large hidden opportunity sets.

14. Current Limitations

P34 is at an early technical-report stage. The benchmark evidence is promising but bounded. The following limitations should be treated as first-order constraints:

- Synthetic evidence. The principal reported benchmarks use synthetic real-world-looking markets. Synthetic tests are useful for isolating partial observability and selection bias, but they do not prove live-market performance.
- Limited regime coverage. The generalized final meta-stage and universe selector have been tested on a limited set of regimes relative to the diversity of real markets.
- Current implementation is not fully differentiable. Boosting-based and materialized-matrix components are used today. This improves debuggability but does not yet realize the intended full neural signal-flow architecture.
- Portfolio unrolling can be expensive. Large menus create dimensionality and compute challenges. Pareto filtering and menu design reduce this cost, but scaling remains an engineering priority.
- Menu quality is decisive. If a business cannot ground actions, sizes, constraints, and outcomes accurately, P34 cannot reliably optimize them.
- Low-liquidity effects are hard. If selecting one deal changes the economic validity of many other deals in an unobserved way, the market may require additional impact modeling before deployment.
- Comparisons are incomplete. The benchmark uses a strong tabular baseline, but it does not exhaustively compare against all possible custom RL, optimization, or quant systems.
- Not a profit guarantee. P34 can reduce certain failure modes but cannot guarantee profitability under adversarial shocks, discontinuous market breaks, execution failures, or invalid data.
- Computable-market identification is not automated. A market must contain enough information to translate into exploitable inefficiency, but P34 does not yet provide a clear computable test for whether a specific market is computable.
- Manipulative and adversarial markets remain risky. This release does not explicitly enter adversarial trading strategies or counter-manipulative actions against other market participants.
- Distant-menu rejection is not yet exposed as an explicit path. If deployment-time menus are far from training context in latent space, the model may not currently refuse to trade automatically.
- The unlabeled menu must exist. The business must historically have had more opportunities than it executed, creating a policy-selection signal. Some problem formulations may lack this signal.
- A menu must exist or be virtually assembled. For case-by-case businesses such as online micro-lending, P34 may require a virtual time-window menu that scores new deals against a rolling portfolio context.
- Regime and policy changes can cause interpretation explosion. Spurious business telemetry changes may increase memory use, require smaller context windows, and reduce performance.
- Latency ranges from under one minute to minutes or hours depending on menu size and configuration, limiting applicability to high-frequency decision-making.
- True generalization remains under study. P34 has been trained and evaluated on a mix of synthetic and actual market data, but the full generalization envelope is not known.
- Inputs are tabular and relationally split into menus and columns. Automatic feature routing is not yet handled, so menu-wise or market-wise features should not be placed into columns without testing.
- Tested menu time resolutions currently include weeks and minutes; behavior outside that range is not yet established.
- Unknown unknowns remain. This is an early report on a new class of models, so additional limitations may be discovered as more markets are explored.

15. Future Development Roadmap

The next milestones should focus on converting the current proof of architecture into a repeatable, audited, scalable platform for computable markets.

Table 11. Recommended P34 development milestones.

Milestone	Description
Differentiable architecture	Replace or augment boosting blocks with a lifted neural architecture capable of end-to-end backpropagation where useful.
Automated universe scaling	Automate generation, Pareto filtering, and selector training across larger universe budgets and broader regime libraries.
Benchmark suite	Build a public or auditor-accessible synthetic benchmark suite covering declining margin, stationary growth, adversarial players, liquidity shocks, delayed feedback, and label scarcity.
Business menu tooling	Create schema validators, menu simulators, leakage tests, and role-specific tooling for business menu managers.
Shadow-mode deployment	Run P34 recommendations alongside existing business decisions and compare realized outcomes without immediate capital exposure.
External signal marketplace	Standardize how market signals, forecasts, LLM-extracted features, and third-party data streams are added, validated, and pruned.
Governance dashboard	Expose regime state, universe choice, false-positive warnings, portfolio concentration, predicted profit, realized profit, and kill-switch status.
Emergent physics queries	Formalize perturbation and counterfactual probes that identify stable learned relationships between market telemetry and economic outcomes.

16. Release Positioning and Pilot Path

P34 should be positioned as an early but important step toward autonomous business decision-making in partially observed markets. The strongest claim is not that P34 forecasts every outcome better than existing models. The claim is that P34 is structured around the right economic problem: constructing portfolios from biased, incomplete business history while explicitly accounting for false positives, market regimes, and deployment-time uncertainty.

For customers, the immediate value proposition is a pilot path: define menus, train context, run shadow portfolios, compare proposed decisions to current business operations, and deploy gradually with caps and controls. For investors, the thesis is that P34 may become a reusable economic decision layer across many computable markets, especially where hidden operational telemetry can be converted into predictable asset-like value.

The release should therefore be evaluated as a decision architecture and pilotable platform, not as a universal profit guarantee. The practical question for a customer is whether its market can be translated into menus with measurable economic feedback and whether P34 can improve the trade-off between false-positive avoidance and opportunity capture under controlled exposure.

17. Conclusion

P34 represents an early implementation of a PARML architecture: a model class that treats economic decision-making as direct regression and optimization over final business outcomes. Its central contribution is not a new isolated predictor, but a decision flow for biased, partially observed operational markets: menu grounding, regime detection, profit prediction, false-positive correction, universe construction, universe selection, and executable portfolio construction.

The benchmark results show the intended behavior. In a difficult market with hidden bias and declining margin, a tuned GBDT-R baseline entered loss-making trades and lost approximately \$219.8k, while P34 stayed profitable. In a stationary/growing market, P34 preserved performance similar to the optimistic baseline. This suggests that profit-directed universe selection and false-positive correction can materially improve robustness without eliminating upside in easier regimes.

The broader thesis is that many markets beyond public securities are becoming computable. Businesses that can convert operational options into structured menus, observe economic feedback, and deploy controlled decision loops may be able to replace fragile analytics-and-rules stacks with adaptive, auditable, profit-directed models. P34 is a step toward that architecture.

Appendix A. Interface Schema and Business Menu Examples

The minimal schema below is intentionally table-oriented. The purpose is to allow business IT teams and data engineers to produce model-ready menus without deep ML specialization.

Table A1. Minimal interface schema.

Field	Description
X_grounded_biased: DataFrame	Historical business telemetry for rows whose outcome is known or safely grounded.
y_grounded_biased: DataFrame or Series	Final economic outcome, usually profit, contribution margin, yield, or recovery.
x_grounded_biased_keys: DataFrame	Unique deal/instrument/product/lead identifiers used for mutual exclusion and sizing.
x_grounded_biased_dates: DataFrame	Decision time, event time, or market-state time used for regimes and forward tests.
x_grounded_biased_bag_ids: DataFrame	Menu, portfolio, vendor, channel, batch, or other decision grouping.
x_historical_biased_choices: DataFrame	Actions selected by the historical policy. Used to estimate observation bias.
X_grounded_deployment_unlabeled: DataFrame	Current candidate menu rows without known outcomes.
x_grounded_deployment_unlabeled_keys/dates/bag_ids: DataFrames	Identifiers and grouping variables for current menus.

Recommended optional columns include naive profit estimates, simple business-rule scores, expected demand, risk flags, price aggressiveness, timing options, capital use, volume, available inventory, external signal values, and execution constraints. Naive predictors are useful not because they are final, but because they give P34 reference signals to correct, reject, or reuse.

Table A2. Business menu examples.

Market	Menu representation
Amazon wholesale	ASIN/vendor/date/quantity/order cost/estimated selling price/fees/sales velocity/inventory constraints.
Moving leads	Lead source/location/time/quoted price/service capacity/probability of acceptance/expected revenue/cancellation risk.
Truck load board	Lane/equipment/date/bid price/estimated cost/deadhead/counterparty/freight attributes/time windows.
Electricity	Node/time/block size/price/forecast demand/weather/grid constraint/risk adjustment.
Prediction markets	Market/outcome/price/size/timing/liquidity/news signal/settlement risk.

Appendix B. Benchmark Methodology

The benchmark design is intentionally closer to business reality than to a clean supervised-learning benchmark. The model is not handed a complete market label matrix. It sees the subset of actions selected or safely grounded by the historical business process and must reason about what the unobserved alternatives imply.

3. Generate a market with latent demand archetypes, costs, weekly demand paths, downside regimes, market difficulty, and hidden business-policy thresholds.
4. Unroll candidate actions over quantity values, producing a full menu of key/date/quantity rows.
5. Apply a biased historical business policy to determine which actions are observed and which are unlabeled.
6. Apply safe grounding where smaller quantities can be inferred from executed larger quantities.
7. Split data by time into historical fit dates and future evaluation dates.
8. Train the baseline regressor and P34 using the historical data available under the partial-observation rules.
9. Evaluate both systems on future menus, aggregating selected key-quantity actions into portfolios.
10. Compare actual portfolio profit, predicted portfolio profit, positive-portfolio success, and efficiency versus the oracle positive upper bound.

The synthetic market includes deliberately broken or noisy features. These features test whether broad feature ingestion and auto-pruning can avoid negative effects from misleading telemetry.

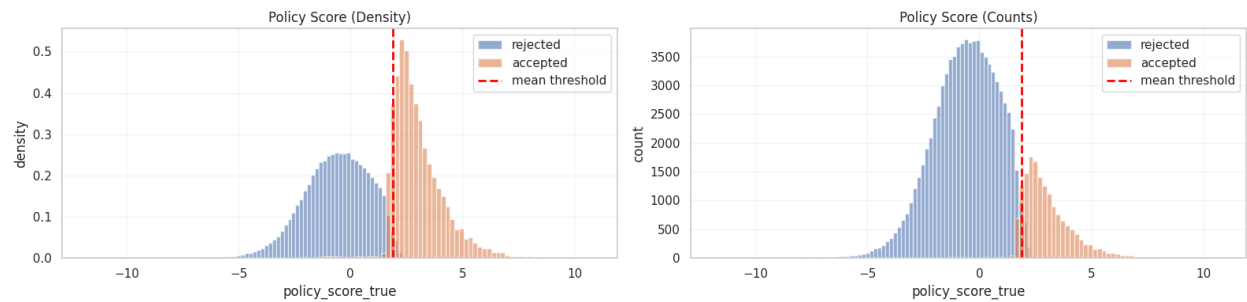


Figure B1. Historical policy score distribution for rejected and accepted deals. The overlap demonstrates why logged business decisions are not clean labels of market truth.

Appendix C. Backtest Methodology

The backtest evaluates each bag independently. A bag is a menu of mutually constrained options available at one evaluation point. The baseline and P34 both operate on the same candidate rows. The baseline selects the maximum positive predicted quantity for each key. P34 returns a portfolio of selected rows after universe selection and false-positive correction.

Actual realized portfolio profit is computed by joining selected key-quantity rows back to the hidden synthetic target. The model does not have access to these target values at inference time. The oracle positive upper bound is calculated only for evaluation and represents the best possible choice if future realized profit were already known.

The backtest notebook also saves per-bag multiverse telemetry. This is important for engineering because aggregate profit alone does not explain why a decision was made. The telemetry allows model developers to inspect the selected universe, false-positive correction, predicted and actual profit, and Pareto features for each bag.

Appendix D. Glossary of Core Terms

Table D1. Glossary.

Term	Definition in this paper
PARML	Profit-As-Regression Machine Learning: a model class trained directly against final economic outcomes rather than intermediate forecasts.
P34	HyperC's current implementation of a PARML decision architecture.
Business menu	A structured set of candidate actions available to a business at a decision point.
Bag	A menu, portfolio, vendor, channel, batch, or other grouping of mutually related candidate actions.
Safe grounding	Label expansion where a known executed quantity safely implies outcomes for smaller quantities under the same context.
False positive	An action that appears profitable under historical or model pattern matching but realizes negative economic outcome.
Universe	A candidate interpretation of historical and future market behavior under partial observability, bias, regime, and drift assumptions.
Universe selector	The meta-level component that chooses among plausible universes according to the policy objective and generalized telemetry.
Latent-space speculation	Estimating economic value where the true market state exists but is not explicitly materialized as clean market data.
Latent-space recovery	Creating rigorous financial dashboard and decision infrastructure from hidden operational telemetry.
Emergent business physics	Stable learned relationships between telemetry and economic outcomes, empirically probed through validation and counterfactual tests.

Appendix E. Source Coverage Map

This integrated paper preserves the information from both input documents while removing redundant repetition and organizing the material as a single technical paper.

Table E1. Source coverage map.

Source content	Integrated location
Industry article thesis and release-candidate positioning	Front matter, Abstract, Executive Summary, Sections 2, 16, and 17.
Industry article problem statement on biased business data and optimistic models	Section 3.

Source content	Integrated location
Industry article applicability scope, latent-space speculation, crypto example, retail market-sizing note	Section 4 and References/Notes.
Industry article business menu definition, special columns, two-context analogy, menu-to-portfolio figure	Section 5 and Figure 1.
Industry article early benchmark evidence and representative benchmark chart/table	Section 9, Figure 8, and Table 8.
Industry article market implications and latent-space recovery	Section 12.
Industry article venture investing analogy	Section 13.
Industry article limitations and release positioning	Sections 14 and 16.
Technical report abstract, executive summary, scope, and introduction	Abstract, Executive Summary, Sections 1 and 2.
Technical report computable markets and business menu interface	Sections 4 and 5, Appendix A.
Technical report P34 model overview and input contract	Sections 5 and 6.
Technical report engineering principles and figures	Section 7 and Figures 3-4.
Technical report synthetic market, bias diagnostics, baseline, backtest mechanics, metrics	Section 8 and Appendices B-C.
Technical report performance benchmarking and interpretation	Section 9 and Figures 7-8.
Technical report competitive landscape	Section 10.
Technical report safety, risk, governance, no-trade, audit requirements	Section 11.
Technical report limitations, roadmap, conclusion, appendices, and references	Sections 14-17, Appendices A-E, and References/Notes.

References and Notes

- [1] BlackRock. "Aladdin: The Language of the Whole Portfolio." <https://www.blackrock.com/aladdin>
- [2] QuantConnect. "Open Source Algorithmic Trading Platform." <https://www.quantconnect.com/>
- [3] SAS. "SAS Intelligent Decisioning." https://www.sas.com/en_us/software/intelligent-decisioning.html
- [4] Pegasystems. "Pega Customer Decision Hub." <https://www.pegasystems.com/products/decision-hub>
- [5] DataRobot. "AI Platform." <https://www.datarobot.com/product/ai-platform/>
- [6] H2O.ai. "H2O AI Cloud." <https://h2o.ai/platform/ai-cloud/>
- [7] Shi, Y. et al. "Kronos: A Foundation Model for the Language of Financial Markets." arXiv:2508.02739, 2025.
- [8] Swaminathan, A. and Joachims, T. "Counterfactual Risk Minimization: Learning from Logged Bandit Feedback." arXiv:1502.02362, 2015.
- [9] Fujimoto, S., Meger, D., and Precup, D. "Off-Policy Deep Reinforcement Learning without Exploration." arXiv:1812.02900, 2018.
- [10] Liu, X.-Y. et al. "FinRL: A Deep Reinforcement Learning Library for Automated Stock Trading in Quantitative Finance." arXiv:2011.09607, 2020.
- [11] Markowitz, H. "Portfolio Selection." The Journal of Finance, 7(1), 77-91, 1952.
- [12] Wu, S. et al. "BloombergGPT: A Large Language Model for Finance." arXiv:2303.17564, 2023.
- [13] Black, F. and Scholes, M. "The Pricing of Options and Corporate Liabilities." Journal of Political Economy, 81(3), 637–654, 1973. doi:10.1086/260062. See also Merton, R. C. "Theory of Rational Option Pricing." Bell Journal of Economics and Management Science, 4(1), 141–183, 1973.
- [14] National Retail Federation forecast as reported by AP News, March 2026: U.S. retail sales forecast to reach approximately \$5.6 trillion in 2026.
- [15] eMarketer/Statista, "Total retail sales worldwide," 2023–2026 (global retail sales projected at approximately \$32.8 trillion for 2026). Estimates vary by provider and methodology; an alternative Statista channel breakdown placed total (online + offline) worldwide retail at roughly \$29 trillion for 2024.
- [16] HyperC internal source material supplied for this integrated paper: P34 technical report and benchmark notebooks, June 2026.
- [17] HyperC source documents supplied for this merge: P34 industry professional article release candidate and P34 industry-quality technical report, June 2026.